

Name and Surname: _____

Student ITS No: _____

Qualification: _____ Year of Study: _____ Semester: _____

Assignment due date: _____ Date submitted: _____

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: _____

Date: _____

Integrated Machine Learning Systems Project

You are appointed as a Machine Learning Consultant to design, implement, and evaluate two advanced intelligent systems using publicly available South African datasets. The objective of this project is to demonstrate your ability to integrate data preprocessing, advanced machine learning techniques, reinforcement learning, large language models, and ethical considerations into fully functional, reproducible systems.

You are required to complete both components described below as part of one comprehensive project submission.

This assignment assesses:

- ❖ Advanced machine learning integration
- ❖ Reinforcement learning modelling
- ❖ Transformer-based NLP implementation
- ❖ Retrieval-augmented generation
- ❖ Ethical reasoning in AI systems
- ❖ Reproducible research practices

COMPONENT A**(50 Marks)****Hybrid Reinforcement Learning and Ensemble Learning System**

Using the **South African Traffic Accident Dataset** <https://www.kaggle.com/datasets/velile/car-accidents>, design and implement a hybrid intelligent system that improves accident prediction and supports adaptive intervention strategies.

System Requirements

Your system must include the following:

1. Data Preparation and Preprocessing (10 Marks)

- ❖ Handle missing values using appropriate imputation techniques.
- ❖ Encode categorical variables (e.g., province, accident type).
- ❖ Detect and treat outliers where appropriate.
- ❖ Justify preprocessing decisions.

2. Ensemble Learning for Accident Prediction (15 Marks)

- Train **at least two ensemble models**, such as:
 - ❖ Random Forest
 - ❖ XGBoost
 - ❖ CatBoost
- Evaluate performance using:
 - ❖ Accuracy
 - ❖ Precision
 - ❖ Recall
 - ❖ F1-score
- Present and interpret a confusion matrix.
- Critically discuss bias–variance trade-offs between the models.

3. Reinforcement Learning for Accident Prevention

(15 Marks)

Model accident prevention as a **Markov Decision Process (MDP)**:

- Define:
 - ❖ States (e.g., road conditions, accident frequency levels)
 - ❖ Actions (e.g., safety campaigns, traffic law enforcement intensity)
 - ❖ Rewards (e.g., reduction in predicted accident severity)
- Implement a **Q-learning agent** to learn optimal safety intervention policies.
- Demonstrate convergence behaviour and policy improvement.

4. System Integration and Evaluation

(10 Marks)

- Integrate ensemble model predictions with the learned RL intervention strategy.
- Provide:
 - ❖ Confusion matrix
 - ❖ Policy evaluation table
- Critically analyse the exploration–exploitation trade-off.
- Discuss practical and ethical implications of automated decision-making in road safety.

Large Language Model (LLM) and Retrieval-Augmented Generation (RAG) System

Using the **South African Parliamentary Hansard Transcripts Dataset**,

[https://www.parliament.gov.za/hansard-papers?sorts\[date\]=-1&page=8&offset=70](https://www.parliament.gov.za/hansard-papers?sorts[date]=-1&page=8&offset=70) design and implement a Large Language Model-based intelligent system for sentiment analysis and policy-focused question answering.

1. Data Preparation and Embedding Generation

(10 Marks)

- Clean and tokenize parliamentary transcripts.
- Apply sub-word tokenization (e.g., BPE or SentencePiece).
- Generate semantic embeddings using a transformer-based model.
- Justify preprocessing and tokenization decisions.

2. Fine-Tuning a Transformer Model

(10 Marks)

- Fine-tune a pre-trained transformer model (e.g., BERT or GPT-2) for:
 - ❖ Policy sentiment classification (positive/negative/neutral)
- Evaluate using:
 - ❖ Accuracy
 - ❖ Precision
 - ❖ Recall
 - ❖ F1-score
- Present and interpret a confusion matrix.

3. Retrieval-Augmented Generation (RAG) Implementation

(10 Marks)

- Build a RAG pipeline that:
 - ❖ Uses FAISS (or equivalent) for vector search
 - ❖ Uses HuggingFace models for text generation
- Demonstrate functionality using example queries such as:
 - ❖ “What was the sentiment on education policy in 2023?”
- Evaluate response relevance and factual grounding.

4. Ethics, Risk, and Responsible AI

(20 Marks)

Provide a critical discussion addressing:

- Bias in accident datasets and parliamentary transcripts.
- Transparency and explainability challenges in ensemble models and LLMs.
- Risks of harmful or misleading generated content.
- Mitigation strategies (e.g., prompt engineering, content filtering, human-in-the-loop systems).
- Responsible AI deployment considerations within the South African context.

Required Deliverables

Students must submit a **single zipped project folder (maximum size: 50MB)** containing ALL of the following:

1. Python Notebook (.ipynb)

- ❖ Complete implementation of both systems.
- ❖ Must execute from top to bottom without errors.
- ❖ Include clear section headings aligned to the assignment structure.
- ❖ Include code, outputs, visualisations, and markdown explanations.

2. Professional PDF Report (3–5 pages)

- ❖ Introduction
- ❖ Methodology
- ❖ Results and Evaluation
- ❖ Discussion and Ethical Considerations
- ❖ Conclusion and Recommendations

3. Dataset

- ❖ Include the pre-processed dataset

4. README.txt

- ❖ Instructions for running the notebook
- ❖ Python version used
- ❖ List of dependencies
- ❖ Expected runtime
- ❖ Hardware requirements (if applicable)

Submissions containing incomplete components or failing to execute correctly will not be awarded marks.

END OF ASSIGNMENT!!!